

Classically perfect blocking

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I. PERTURBATIVELY OPTIMIZED PERFECT BLOCKING FOR THE WILSON GAUGE ACTION

A. Goal and philosophy

The usual perfect-action program starts from a prescribed renormalization-group (RG) transformation and determines the corresponding fixed-point (FP) action [1, 2]. For gauge fields, this program was developed in detail in Ref. [3], where the blocking transformation itself was generalized in order to improve locality and rotational properties of the FP action. Here we reverse the question.

Our goal is to keep the fine and coarse actions fixed to the Wilson plaquette action and instead optimize the blocking transformation:

$$S_W[U] \xrightarrow{\mathcal{B}} S_W[U^{(c)}]. \quad (1.1)$$

The important structural restriction, motivated by the inverse-blocking application, is that blocking is performed in two distinct steps¹:

$$U_\mu(x) \longrightarrow V_\mu(x) \longrightarrow U_\mu^{(c)}(X) = V_\mu(2X)V_\mu(2X + \hat{\mu}). \quad (1.2)$$

Thus $V_\mu(x)$ is a genuine smoothed configuration on the original fine lattice. This is the configuration that will eventually be reconstructed in the inverse RG/upscaling procedure. Volume reduction occurs only in the second step.

We use a deterministic (hard) blocking. There is no Gaussian “soft blocking” term. This choice is also motivated by our experience in scalar models, where a stochastic/soft blocking relation is significantly harder for a machine-learning inverse map to reproduce accurately².

B. From the exact RG transformation to hard blocking

The blocked action is defined by the exact renormalization-group transformation

$$e^{-S'(V)} = \int DU e^{-S(U) - T(U, V)}, \quad (1.3)$$

where U denotes fine links, V coarse links, and $T(U, V)$ is the blocking kernel.

For gauge theories it is useful to separate the overall coupling,

$$S(U) = \beta \mathcal{A}(U), \quad T(U, V) = \beta \mathcal{T}(U, V). \quad (1.4)$$

In the classical limit $\beta \rightarrow \infty$, the functional integral is dominated by the minimum of the exponent,

$$\mathcal{A}'(V) = \min_U [\mathcal{A}(U) + \mathcal{T}(U, V)]. \quad (1.5)$$

At a fixed point this becomes

$$\mathcal{A}^{\text{FP}}(V) = \min_U [\mathcal{A}^{\text{FP}}(U) + \mathcal{T}(U, V)]. \quad (1.6)$$

We are interested in a deterministic blocking transformation

$$V = \mathcal{B}(U). \quad (1.7)$$

A soft blocking kernel may be written schematically as

$$\mathcal{T}_\kappa(U, V) = \kappa d^2(V, \mathcal{B}(U)), \quad (1.8)$$

¹ [AH: The original perfect action defined a blocking kernel that connected distance-2 sites and parametrized the loops in the kernel that all extended distance-2 away.]

² [AH: This was my experience in the O(3) model, but might not be universal. We can revisit this question if necessary.]

where $d^2 \geq 0$ measures the mismatch between the prescribed coarse field V and the blocked fine field $\mathcal{B}(U)$.

In the hard-blocking limit $\kappa \rightarrow \infty$, configurations with

$$\mathcal{B}(U) \neq V \quad (1.9)$$

receive an infinite penalty. Equation (1.5) therefore reduces to

$$\boxed{S'(V) = \min_{U: \mathcal{B}(U)=V} S(U)} \quad (1.10)$$

This is the form used below in the perturbative calculation.

C. From the perturbative fixed-point equation to nonlinear perfect blocking

The classical fixed-point construction for lattice gauge theories was developed in several stages. It is useful to separate these steps, since they also provide a natural roadmap for the present “inverse” problem in which the action is held fixed and the blocking transformation is optimized.

a. Quadratic order. The first step is the weak-field expansion

$$U_\mu(x) = \exp[iA_\mu(x)] = 1 + iA_\mu(x) - \frac{1}{2}A_\mu^2(x) + \mathcal{O}(A_\mu^3), \quad (1.11)$$

keeping terms in the action up to quadratic order in the gauge field. At this level the fixed-point equation becomes a Gaussian problem and can be solved analytically in momentum space. The blocking transformation is represented by a linear kernel $\Omega_{\mu\nu}(k)$, and the fixed-point condition becomes a recursion relation for the quadratic action kernel, or equivalently for the free propagator. In the classic perfect-action construction, the parameters entering the RG transformation were optimized primarily to make the resulting fixed-point action as local as possible.

This is the level at which we begin our present construction, but with the logic reversed. We keep the quadratic Wilson action fixed and optimize the smearing/blocking kernel so that

$$S_W^{(2)} \xrightarrow{\mathcal{B}} S_W^{(2)} \quad (1.12)$$

as accurately as possible over the entire Brillouin zone.

b. Cubic order. The next step in the traditional fixed-point program is to expand the fixed-point equation to cubic order in the gauge field. The quadratic solution determines the propagator entering this calculation, while the $\mathcal{O}(A^3)$ terms probe the first genuinely non-Abelian structure. In particular, path orderings which are indistinguishable in the quadratic approximation can first be separated at this order.

This will be important for the present construction. For example, several of the one-link paths entering our generalized smearing have identical linearized kernels even though they are distinct $SU(3)$ transporters. The quadratic optimization therefore determines only particular combinations of their coefficients. A cubic calculation would provide the first perturbative criterion for splitting these degeneracies.

D. Nonperturbative extension by ensemble matching

At finite coupling we do not plan to extend the classical fixed-point minimization to arbitrary coarse configurations. Instead, we use the fact that the desired RG transformation should map two equilibrium Wilson ensembles whose lattice spacings differ by a factor of two.

Let β_c denote the bare coupling on the coarse lattice and β_f the coupling for which the Wilson action has lattice spacing

$$a_f = \frac{a_c}{2}. \quad (1.13)$$

We generate independent equilibrium ensembles

$$U_f \sim P_{\beta_f}, \quad U_c \sim P_{\beta_c}, \quad (1.14)$$

and apply the parametrized blocking transformation $\mathcal{B}_\theta$ to the fine configurations,

$$V_c = \mathcal{B}_\theta(U_f). \quad (1.15)$$

The nonperturbative optimization problem is then to choose the smearing parameters θ so that the blocked fine ensemble reproduces the coarse Wilson ensemble. Formally, the desired condition is

$$\mathcal{B}_{\theta*} P_{\beta_f} \simeq P_{\beta_c}, \quad (1.16)$$

where $\mathcal{B}_{\theta*}$ denotes the push-forward of the fine-lattice probability distribution.

In practice, we impose this condition through a set of gauge-invariant observables O_i and minimize an objective of the form

$$\mathcal{L}(\theta) = \sum_i w_i \left[\langle O_i \rangle_{\beta_c} - \langle O_i(\mathcal{B}_\theta[U_f]) \rangle_{\beta_f} \right]^2. \quad (1.17)$$

The operator basis can include Wilson loops of several sizes and shapes, gradient-flow observables, and longer-distance quantities sensitive to the effective lattice spacing. The perturbative perfect-blocking calculation provides the weak-coupling starting point and a compact parametrization of $\mathcal{B}_\theta$, while the finite-coupling ensemble matching determines the nonlinear corrections relevant for the actual Wilson ensembles.

This approach avoids the need to solve the classical fixed-point equation configuration by configuration. Instead, it directly targets the property needed for inverse blocking:

$$\text{fine Wilson ensemble at } \beta_f \xrightarrow{\mathcal{B}_\theta} \text{coarse Wilson ensemble at } \beta_c. \quad (1.18)$$

II. QUADRATIC APPROXIMATION

A. General one-link smearing

We first form a gauge-covariant sum of paths with the same endpoints as the original link,

$$Q_\mu(x) = c_0 U_\mu(x) + \sum_a c_a \sum_{p \in \mathcal{O}_a(\mu)} U(p; x \rightarrow x + \hat{\mu}), \quad (2.1)$$

where $\mathcal{O}_a(\mu)$ denotes a complete orbit under the lattice symmetries that leave the directed link (x, μ) invariant. The overall normalization of Q_μ is immaterial before projection. The smeared link is

$$V_\mu(x) = \mathcal{P}_{SU(3)}[Q_\mu(x)]. \quad (2.2)$$

The projection may be defined, for example, as the $SU(3)$ matrix maximizing $\text{Re Tr } V_\mu Q_\mu^\dagger$, or equivalently through polar decomposition followed by removal of the determinant, [or more likely by stout smearing construction](#).

Hard projection should not be confused with stout smearing [4]. Stout smearing is an analytic exponential update and is not the same nonlinear map as projection of a nonunitary path sum. However, the two constructions can be matched at linear order around the identity. Consequently, for the perturbative FP calculation below, only the linearized smearing kernel matters; the distinction between projection and stout realization enters beyond quadratic order.

A useful short-path basis is

$$\begin{aligned} P_0 : & \quad \mu, \\ P_1 : & \quad \nu \mu (-\nu), \\ P_2 : & \quad \nu \nu \mu (-\nu) (-\nu), \\ P_3 : & \quad \nu \rho \mu (-\rho) (-\nu), \\ P_4 : & \quad \nu \rho \mu (-\nu) (-\rho), \\ P_5 : & \quad \nu \mu \mu (-\nu) (-\mu), \\ P_6 : & \quad (-\mu) \nu \mu \mu (-\nu), \end{aligned} \quad (2.3)$$

with all symmetry-related orientations included. P_5 and P_6 are distinct paths but symmetry requires a common coefficient. Likewise P_3 and P_4 are distinct non-Abelian paths, although they become degenerate at [linear order](#) after completion of the full hypercubic orbit.

B. Linearized one-link kernel

Write

$$U_\mu(x) = \exp[iA_\mu(x)]. \quad (2.4)$$

To first order,

$$V_\mu(k) = 1 + iA_\mu^{(s)}(k) + O(A^2), \quad A_\mu^{(s)}(k) = \omega_{\mu\nu}(k)A_\nu(k). \quad (2.5)$$

If the unprojected path sum satisfies

$$Q_\mu(x) = q_0 \mathbf{1} + i \mathcal{W}_{\mu\nu} A_\nu + O(A^2), \quad (2.6)$$

then hard projection gives

$$\omega_{\mu\nu}(k) = \frac{W_{\mu\nu}(k)}{q_0}. \quad (2.7)$$

The division by q_0 is essential. It guarantees that a pure-gauge mode is mapped into the same pure-gauge mode:

$$\omega_{\mu\nu}(k) \hat{k}_\nu = \hat{k}_\mu, \quad (2.8)$$

with the appropriate lattice momentum convention.

For the short-path basis above, the zero-field normalization is

$$q_0 = 1 + 6K_1 + 6K_2 + 24K_3 + 24K_4 + 12K_5. \quad (2.9)$$

At linear order the two diagonal five-link classes satisfy

$$W_{\mu\nu}^{(3)}(k) = W_{\mu\nu}^{(4)}(k), \quad (2.10)$$

so only

$$K_{34} = K_3 + K_4 \quad (2.11)$$

is perturbatively measurable. Their difference is intrinsically non-Abelian and first enters beyond the quadratic approximation.

C. Linearized two-link blocking

Let the linearized smeared fine-link field be

$$A_\mu^{(s)}(k) = \omega_{\mu\nu}(k)A_\nu(k), \quad (2.12)$$

where $\omega_{\mu\nu}(k)$ is the one-link smearing kernel.

The coarse link is defined as the product of two neighboring smeared fine links,

$$U_\mu^{(c)}(X) = V_\mu(2X)V_\mu(2X + \hat{\mu}). \quad (2.13)$$

Writing

$$U_\mu^{(c)}(X) = 1 + iB_\mu(X) + O(A^2), \quad (2.14)$$

one obtains at linear order

$$B_\mu(X) = A_\mu^{(s)}(2X) + A_\mu^{(s)}(2X + \hat{\mu}). \quad (2.15)$$

To see the corresponding momentum-space relation, consider a single fine-lattice Fourier mode,

$$A_\mu^{(s)}(x) = e^{ik \cdot x} A_\mu^{(s)}(k). \quad (2.16)$$

Then

$$A_\mu^{(s)}(2X) = e^{2ik \cdot X} A_\mu^{(s)}(k), \quad (2.17)$$

while

$$A_\mu^{(s)}(2X + \hat{\mu}) = e^{2ik \cdot X} e^{ik_\mu} A_\mu^{(s)}(k). \quad (2.18)$$

Therefore

$$B_\mu(X) = e^{2ik \cdot X} (1 + e^{ik_\mu}) A_\mu^{(s)}(k). \quad (2.19)$$

Because momenta that differ by π in any fine-lattice direction become identical after the factor-two decimation, a given coarse momentum K receives contributions from the sixteen fine-lattice momenta

$$k_l = \frac{K}{2} + \pi l, \quad l_\alpha = 0, 1. \quad (2.20)$$

We define the **complete fine-to-coarse linear blocking kernel** $\Omega_{\mu\nu}(k)$ by

$$B_\mu(K) = \frac{1}{16} \sum_{l_\alpha=0,1} \Omega_{\mu\nu}(k_l) A_\nu(k_l), \quad k_l = \frac{K}{2} + \pi l. \quad (2.21)$$

The factor $1/16$ depends on the normalization convention for the Fourier transform; it may equivalently be absorbed into the definition of $\Omega_{\mu\nu}$.

Using

$$A_\mu^{(s)}(k) = \omega_{\mu\nu}(k) A_\nu(k) \quad (2.22)$$

in Eq. (2.19), the complete blocking kernel is

$$\boxed{\Omega_{\mu\nu}(k) = (1 + e^{ik_\mu}) \omega_{\mu\nu}(k)}. \quad (2.23)$$

Thus there are two distinct kernels in the construction:

$$A_\nu(k) \xrightarrow{\omega} A_\mu^{(s)}(k) \xrightarrow{\text{two-link product}} B_\mu(K), \quad (2.24)$$

where $\omega_{\mu\nu}$ describes only the one-link smearing and $\Omega_{\mu\nu}$ describes the complete fine-to-coarse blocking.

The factor multiplying $\omega_{\mu\nu}$ can also be written as

$$1 + e^{ik_\mu} = 2e^{ik_\mu/2} \cos\left(\frac{k_\mu}{2}\right). \quad (2.25)$$

If gauge fields are Fourier transformed using the natural midpoint of the link, $x + \hat{\mu}/2$, the phase $e^{ik_\mu/2}$ is absorbed into the Fourier convention. In this convention the complete blocking kernel becomes

$$\boxed{\Omega_{\mu\nu}(k) = 2 \cos\left(\frac{k_\mu}{2}\right) \omega_{\mu\nu}(k)}. \quad (2.26)$$

The factorized form

$$\Omega(k) = D(k) \omega(k), \quad (2.27)$$

with

$$D_{\mu\nu}(k) = 2 \cos\left(\frac{k_\mu}{2}\right) \delta_{\mu\nu} \quad (2.28)$$

in the midpoint convention, is a defining restriction of the present blocking transformation. The coarse link is not an arbitrary transporter from x to $x + 2\hat{\mu}$; it is required to be the product of two identical smeared one-link transporters.

D. Hard-block Gaussian RG equation

Consider a quadratic gauge action

$$S[A] = \frac{1}{2} \sum_k A_\mu(-k) \rho_{\mu\nu}(k) A_\nu(k). \quad (2.29)$$

For a deterministic blocking $B = \mathcal{B}A$, the blocked action is

$$S'[B] = \min_{A: \mathcal{B}A=B} S[A]. \quad (2.30)$$

After temporary gauge fixing, or equivalently after restricting to the transverse subspace, let

$$G(k) = \rho^{-1}(k). \quad (2.31)$$

For scale factor $b = 2$, the coarse momentum K has $2^4 = 16$ aliases

$$k_l = \frac{K}{2} + \pi l, \quad l_\alpha = 0, 1, \quad (2.32)$$

and the blocked covariance is

$$G'(K) = \frac{1}{16} \sum_{l \in \{0,1\}^4} \Omega(k_l) G(k_l) \Omega^\dagger(k_l). \quad (2.33)$$

This is the hard-block analogue of the perturbative FP recursion used in Ref. [3]; the additive $1/\kappa$ term associated with a soft Gaussian blocking kernel is absent.

For the Wilson plaquette action, the transverse free propagator may be written as

$$G_{\mu\nu}^W(k) = \frac{1}{\widehat{k}^2} \left[\delta_{\mu\nu} - \frac{\widehat{k}_\mu \widehat{k}_\nu^*}{\widehat{k}^2} \right], \quad \widehat{k}^2 = \sum_\alpha |\widehat{k}_\alpha|^2. \quad (2.34)$$

The perturbative perfect-blocking condition is therefore

$$\boxed{G^W(K) = \frac{1}{16} \sum_l D(k_l) \omega(k_l) G^W(k_l) \omega^\dagger(k_l) D^\dagger(k_l)} \quad (2.35)$$

with

$$D_{\mu\nu}(k) = \delta_{\mu\nu} (1 + e^{ik_\mu}). \quad (2.36)$$

We optimize the complete Brillouin zone rather than only the small-momentum expansion. A convenient objective is

$$\mathcal{E} = \left\langle \frac{\|P_T(K) [G^W(K) - G'(K)] P_T(K)\|_F^2}{\|G^W(K)\|_F^2} \right\rangle_{K \in \text{BZ}}, \quad (2.37)$$

where P_T is the transverse projector. Gauge covariance of each path orbit is checked independently and is not imposed as a numerical fit constraint.

[AH: This minimization is equivalent to minimizing the difference of all distinct operator expectation values. Since this is a Gaussian approximation, we can do it in Fourier space and do it analytically. But the idea is the same.]

E. Preliminary exploration of the path space

A systematic enumeration of symmetry-completed one-link paths shows a strong collapse of the geometrical path space after linearization. In the current exploratory implementation, the dimension of the independent perturbative kernel space was found to grow approximately as

$$\dim \mathcal{W}_{\ell \leq 5} = 5, \quad \dim \mathcal{W}_{\ell \leq 7} = 12, \quad \dim \mathcal{W}_{\ell \leq 9} = 23. \quad (2.38)$$

Thus thousands of geometrically distinct paths at length nine correspond to only a few dozen independent quadratic kernels.

The short-path basis through length five gives a full-Brillouin-zone residual of order 10^{-3} . Adding the independent length-seven directions lowers the residual by roughly one to two orders of magnitude. Extending through length nine lowers it further to the 10^{-5} range in exploratory full-zone tests. One representative convergence table from the present prototype is

maximum path length	fit grid	$\mathcal{E}$
5	6^4	1.7×10^{-3}
7	6^4	9.1×10^{-5}
9	6^4	5.8×10^{-6}
5	8^4 validation	1.8×10^{-3}
7	8^4 validation	1.1×10^{-4}
9	8^4 validation	1.5×10^{-5}

These numbers should be viewed as preliminary until the production implementation is rerun with a fixed Fourier convention and the compact parametrization below. The robust qualitative result is the rapid reduction of the Wilson FP residual as the one-link smearing space is enlarged; no clear nonzero floor has appeared through length nine.

[AH: By increasing the blocking kernel we test if the Wilson action could be a FP action for some RGblock transformation. My feeling is that only if we allow an infinite range smearing. But the observation that $\mathcal{E}$ drops rapidly and in the 10^{-5} range with length-7 blocking is good enough: using a hard blocking and not too large smoothing kernels we can get the Wilson action to be very close to a FP action. I don't think we need to put for more. The real test will be the finite lattice spacing comparison in any case.]

F. Compact parametrization: a polynomial covariant link smoother

For practical inverse blocking we do not want tens or hundreds of unrelated path coefficients. A more natural parametrization is obtained by generating the path shells from a single gauge-covariant nearest-neighbor link operator.

For a link-valued field $X_\mu(x)$ with the same gauge transformation law as $U_\mu(x)$, define the transverse covariant link Laplacian

$$(\mathcal{L}X)_\mu(x) = \sum_{\nu \neq \mu} \left[U_\nu(x) X_\mu(x + \hat{\nu}) U_\nu^\dagger(x + \hat{\mu}) + U_\nu^\dagger(x - \hat{\nu}) X_\mu(x - \hat{\nu}) U_\nu(x - \hat{\nu} + \hat{\mu}) - 2X_\mu(x) \right]. \quad (2.39)$$

When $X = U$, the first application of $\mathcal{L}$ generates the usual three-link staples. Repeated applications generate symmetry-correlated one-link paths of length five, seven, nine, $\dots$, while preserving the endpoints $(x, x + \hat{\mu})$.

We therefore propose the compact ansatz

$$Q_\mu = \left[1 + a_1 \mathcal{L} + a_2 \mathcal{L}^2 + a_3 \mathcal{L}^3 + a_4 \mathcal{L}^4 \right] U_\mu \quad (2.40)$$

followed by a single hard projection

$$V_\mu = \mathcal{P}_{SU(3)} Q_\mu. \quad (2.41)$$

Degree four produces paths through length nine, precisely the range for which the unrestricted perturbative basis already gives a residual near 10^{-5} .

Equation (2.40) is the minimal compact parametrization. It has several advantages:

1. only four nontrivial parameters are required;
2. every term is gauge covariant by construction;
3. all hypercubic symmetry partners occur automatically;
4. longer paths are generated with correlated coefficients rather than being fitted independently;
5. the output is still a one-link field on the original fine lattice;

Maximum path length	$\dim \mathcal{W}$	$\mathcal{E}$ on 6^4 fit grid	$\mathcal{E}$ on 8^4 validation grid
5	5	1.7×10^{-3}	1.8×10^{-3}
7	12	9.1×10^{-5}	1.1×10^{-4}
9	23	5.8×10^{-6}	1.5×10^{-5}

TABLE I. Preliminary full-Brillouin-zone optimization of the perturbative perfect-blocking transformation. $\mathcal{W}$ denotes the space of independent linearized one-link smearing kernels generated by all symmetry-completed paths up to the indicated maximum length. The objective $\mathcal{E}$ is the normalized squared difference between the Wilson propagator and the propagator obtained after optimized smearing followed by factor-two blocking. Parameters were optimized on a uniform 6^4 momentum grid; the final column evaluates the same parameters on an independent 8^4 grid without refitting.

Coefficient	Representative path	Perturbative role
K_1	$\nu \mu (-\nu)$	nearest transverse staple
K_2	$\nu \nu \mu (-\nu) (-\nu)$	extended planar staple
$K_{34} = K_3 + K_4$	$\nu \rho \mu (-\rho) (-\nu)$	two-transverse-direction paths
K_5	$\nu \mu \mu (-\nu) (-\mu)$	longitudinal hook

TABLE II. Independent parameters of the linearized one-link smearing kernel through path length five. The two non-Abelian orderings associated with K_3 and K_4 have identical kernels at linear order, so only their sum K_{34} can be determined by the quadratic perfect-blocking condition.

6. the same construction can be implemented directly and efficiently in the nonlinear $SU(3)$ code.

The unrestricted path study indicates that one additional structure may be needed beyond a pure polynomial in $\mathcal{L}$, because the short-path basis contains an independent “longitudinal hook” direction exemplified by

$$\nu \mu \mu (-\nu) (-\mu). \quad (2.42)$$

A compact extension is therefore

$$Q_\mu = [1 + a_1 \mathcal{L} + a_2 \mathcal{L}^2 + a_3 \mathcal{L}^3 + a_4 \mathcal{L}^4] U_\mu + [b_0 + b_1 \mathcal{L} + b_2 \mathcal{L}^2] (\mathcal{H}U)_\mu, \quad (2.43)$$

where $\mathcal{H}$ denotes the symmetry-completed longitudinal-hook operator whose shortest representative is $P_5 + P_6$. This gives a seven-parameter family. It should be flexible enough to reproduce most of the 23-dimensional length- ≤ 9 perturbative optimum while remaining simple enough for a nonlinear implementation.

A still more conservative version is to retain two independently generated geometrical operators, $\mathcal{L}$ and $\mathcal{H}$, and fit all monomials with total degree at most four,

$$Q_\mu = \left[1 + \sum_{m+n \leq 4} c_{mn} \mathcal{L}^m \mathcal{H}^n \right] U_\mu, \quad (2.44)$$

dropping redundant combinations after a linear-rank analysis. In practice we expect that considerably fewer than the full set of coefficients will be needed.

[AH: Table II gives representative values for the length-5 blocking. The length-7 and 9 kernels have to be parametrized first with independent loops before meaningful comparison can be made. Take all this as preliminary first tests.]

The strong reduction in $\mathcal{E}$ as the path basis is enlarged suggests that there is no obvious residual floor at the level reached so far. Increasing the maximum path length from five to seven reduces the full-Brillouin-zone mismatch by approximately an order of magnitude, and inclusion of length-nine paths gives a further substantial reduction. The close agreement between the fit and validation grids indicates that this improvement is not simply an overfit to the discrete momentum set.

An important feature is the strong reduction of the path space after linearization. Although the number of geometrically distinct gauge paths grows rapidly with path length, only 5, 12, and 23 independent quadratic kernels remain through lengths 5, 7, and 9, respectively. This motivates the search for a much more compact symmetry-adapted parametrization of the smearing kernel.

G. Recommended perturbative program

The next perturbative calculation should proceed in the following order.

1. Construct the exact momentum-space kernels $L_{\mu\nu}(k)$ and $H_{\mu\nu}(k)$ corresponding to $\mathcal{L}$ and $\mathcal{H}$.
2. Optimize Eq. (2.37) first for the four-parameter polynomial (2.40).
3. Add the hook sector in Eq. (2.43) and quantify how much of the unrestricted length- ≤ 9 improvement is recovered.
4. Repeat on progressively denser Brillouin-zone grids and on an independent random/Sobol momentum sample.
5. Examine the pointwise residual, not only the mean objective, with special attention to corners and edges of the Brillouin zone.
6. Only after the perturbative compact kernel is stable, split perturbatively degenerate non-Abelian paths such as P_3 and P_4 and optimize them with nonlinear Monte Carlo matching observables.

The central question is whether the compact seven-parameter family can retain the $\sim 10^{-5}$ quadratic residual of the unrestricted length-nine kernel. If so, it provides a practical candidate for the actual inverse-RG blocking transformation.

H. Interpretation

The emerging picture is conceptually different from the standard perfect-action construction. Rather than moving the complexity into a long-ranged FP action, we attempt to move it into a local but optimized smoothing transformation:

$$\boxed{\text{Wilson action} + \text{optimized one-link smoothing} + \text{exact two-link decimation.}} \quad (2.45)$$

At quadratic order, the numerical evidence so far suggests that the Wilson action can be made very nearly fixed under such a transformation. The factorized construction is particularly attractive for inverse blocking: the first stage produces a physically meaningful smooth fine-lattice configuration, and the second stage is a simple exact reduction of lattice volume.

The perturbative calculation by itself does not establish an exact nonperturbative FP transformation. Non-Abelian distinctions invisible at $O(A^2)$, the nonlinear hard projection, and finite-coupling effects must all be tested. Nevertheless, the rapid decrease of the full-Brillouin-zone quadratic residual as the one-link path space is enlarged suggests that this is a promising direction for a practical “perfect blocking” tailored to the Wilson gauge action.

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